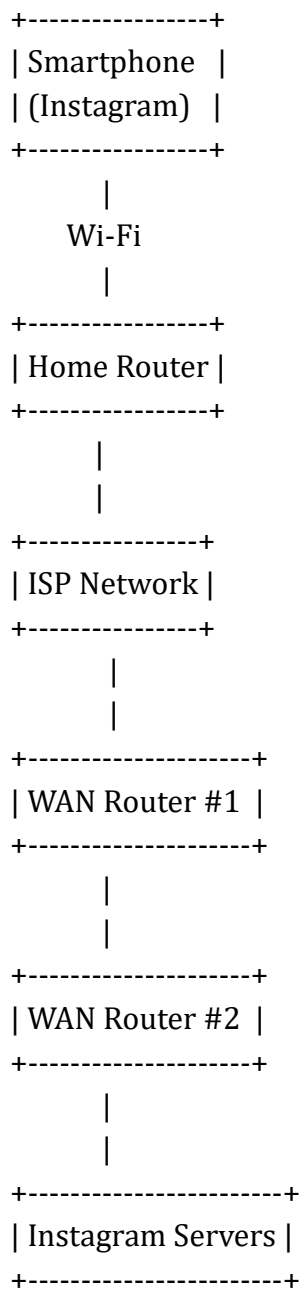


## Wide Area Networks

1. Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

ANS :

- **Network Diagram**



- **Explanation**

1. The smartphone connects to the home Wi-Fi router.
2. The router sends data to the Internet Service Provider (ISP).
3. The data travels through multiple WAN routers across the internet.
4. The request finally reaches Instagram's servers.
5. Instagram sends the requested content back through the same WAN path.

**2. Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).**

**ANS :**

| <b>WAN Technology</b>                | <b>Speed</b>   | <b>Cost</b>    | <b>Real-Life Usage</b>                                         |
|--------------------------------------|----------------|----------------|----------------------------------------------------------------|
| MPLS (Multiprotocol Label Switching) | High           | Expensive      | Large organizations such as Zomato offices connecting branches |
| Leased Line                          | Very High      | Very Expensive | Banks, IRCTC ticketing systems, data centers                   |
| Broadband Internet                   | Medium to High | Affordable     | Home internet, small offices, students                         |

- **Summary**

- MPLS provides reliable and secure communication between company branches.
- Leased Lines offer dedicated bandwidth and excellent performance.
- Broadband Internet is cost-effective and commonly used in homes.

**3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop. Hint : Focus on how the VPN creates a private tunnel over the public WAN.**

**ANS :**

A Virtual Private Network (VPN) allows users to securely access the internet over a public Wide Area Network (WAN).

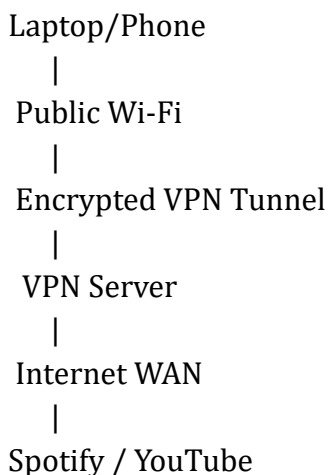
- **Working of VPN**

1. A user connects to a public Wi-Fi network, such as in a coffee shop.
2. The VPN application creates an encrypted tunnel between the user's device and the VPN server.
3. All internet traffic travels through this secure tunnel.
4. Hackers on the same public Wi-Fi cannot read the encrypted data.
5. The VPN server forwards the request to websites such as Spotify or YouTube.
6. Responses return through the same encrypted tunnel.

- **Benefits**

- Protects sensitive information.
- Prevents data theft on public Wi-Fi.
- Hides the user's actual IP address.
- Provides secure remote access over WAN networks.

- **Simple VPN Diagram**



**4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.**

**ANS :**

- **Scenario**

Flipkart wants to connect warehouses located in:

- Ahmedabad
- Mumbai
- Delhi

- **Recommended WAN Technology: MPLS**

**Reason 1: Reliable Connectivity**

MPLS provides stable and reliable communication between different warehouse locations. Important business applications such as inventory management and order processing can run without interruptions.

**Reason 2: Better Security and Performance**

MPLS routes traffic efficiently and securely across the provider's network. It offers lower latency and better performance than standard internet connections, making it ideal for business-critical operations.